NFL_playersalary_projectcode.R

hsoha

2026-05-02

```
# REQUIRED PACKAGES
library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.5.2

## Warning: package 'ggplot2' was built under R version 4.5.2

## Warning: package 'dplyr' was built under R version 4.5.2

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.1.5
## ✓ forcats   1.0.0     ✓ stringr   1.5.1
## ✓ ggplot2   4.0.1     ✓ tibble    3.3.0
## ✓ lubridate 1.9.4     ✓ tidyr     1.3.1
## ✓ purrr     1.1.0
## ── Conflicts ────────────────────────────────────────
tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(car)

## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData

## Warning: package 'carData' was built under R version 4.5.2

##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some

library(ggcorrplot)

## Warning: package 'ggcorrplot' was built under R version 4.5.2
```

```r
library(lmtest)
```

```
## Warning: package 'lmtest' was built under R version 4.5.2
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.5.2
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(ggplot2)
library(ggfortify)
```

```
## Warning: package 'ggfortify' was built under R version 4.5.2
```

```r
library(gridExtra)
```

```
## Warning: package 'gridExtra' was built under R version 4.5.2
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(dplyr)
library(ggplot2)
library(effects)
```

```
## Warning: package 'effects' was built under R version 4.5.3
```

```
## lattice theme set by effectsTheme()
## See ?effectsTheme for details.
```

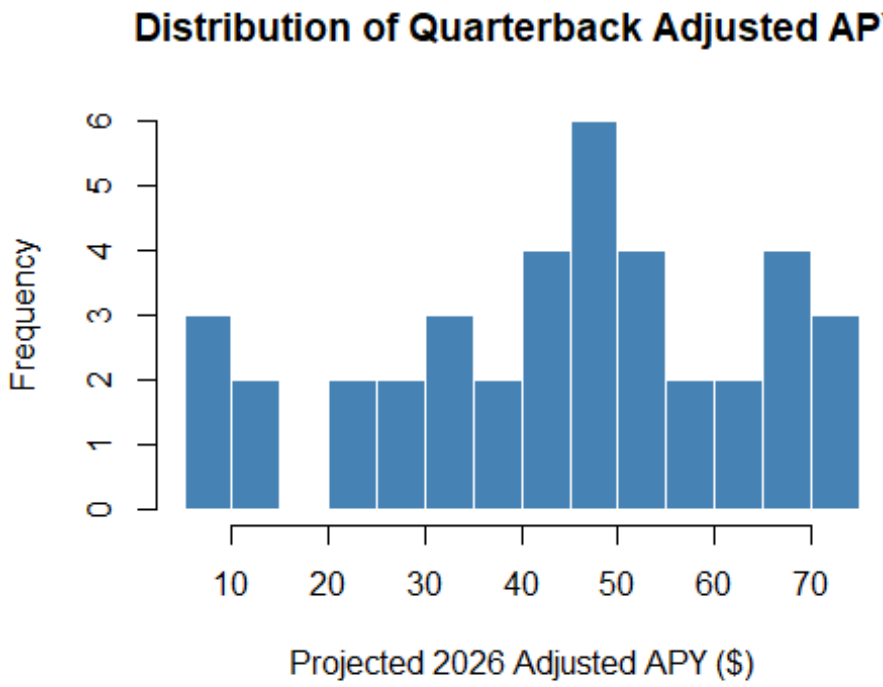
```r
library("tinytex")
```

```
## Warning: package 'tinytex' was built under R version 4.5.3
```

```r
# EDA SECTION
# ========================================

# QUARTERBACK EDA
qb_data <- read.csv("C:/R/STAT 7210/Project/qb_data_final.csv")
qb_data <- qb_data |>
  filter(!is.na(interceptions), !is.na(age_at_contract),
         !is.na(cpoe), !is.na(bmi))
```

```r
# EDA CONDUCTED
hist(qb_data$apy_adj_2026,
     main = "Distribution of Quarterback Adjusted APY",
     xlab = "Projected 2026 Adjusted APY ($)",
     col = "steelblue",      # Professional blue fill
     border = "white",       # Clean white borders between bars
     breaks = 12)            # Adjusts the number of bins
```

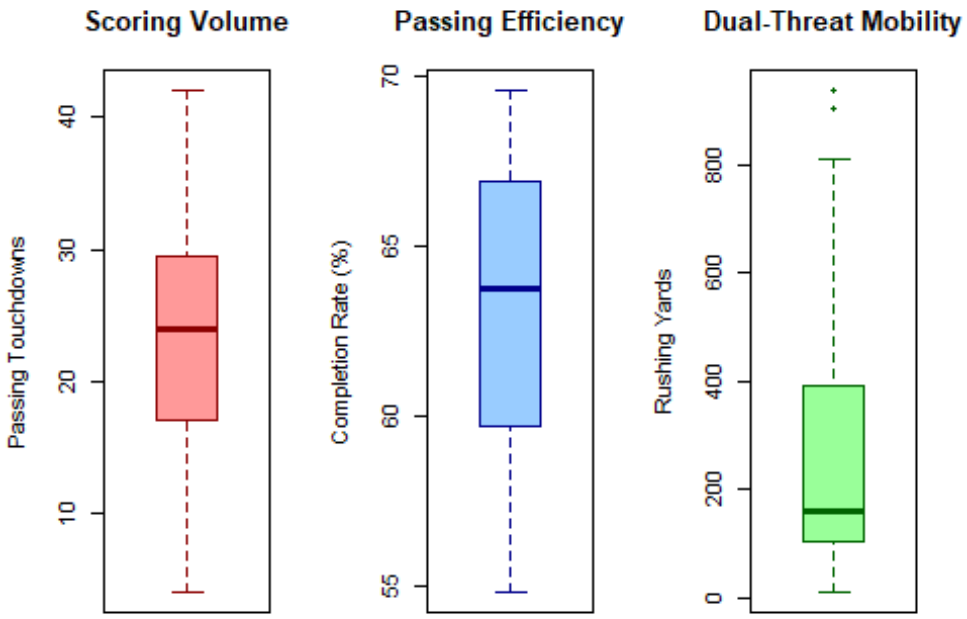



```r
# Set up a 1 row by 3 column grid for the boxplots to save space
par(mfrow = c(1, 3))

# Boxplot 1: Passing TDs
boxplot(qb_data$passing_tds,
        main = "Scoring Volume",
        ylab = "Passing Touchdowns",
        col = "#ff9999",     # Light red fill
        border = "darkred",  # Dark red outline
        pch = 16)            # Solid dots for outliers

# Boxplot 2: Completion Rate
boxplot(qb_data$comp_rate,
        main = "Passing Efficiency",
        ylab = "Completion Rate (%)",
        col = "#99ccff",     # Light blue fill
        border = "darkblue", # Dark blue outline
        pch = 16)
```

```r
# Boxplot 3: Rushing Yards
boxplot(qb_data$rushing_yards,
        main = "Dual-Threat Mobility",
        ylab = "Rushing Yards",
        col = "#99ff99",     # Light green fill
        border = "darkgreen",# Dark green outline
        pch = 16)
```




```r
par(mfrow = c(1, 1))

# subset numeric columns

numeric_cols <- qb_data |>
  select(where(is.numeric)) |>
  select(-any_of(c("player_id", "prior_year", "birth_year", "cap_value")))

numeric_cols <- numeric_cols[, colSums(is.na(numeric_cols)) <
nrow(numeric_cols)]
cor_matrix <- cor(numeric_cols, use = "pairwise.complete.obs")
cor_matrix[is.na(cor_matrix)] <- 0

cat("\nTop Predictors for Adjusted APY (N =", nrow(qb_data), "):\n")

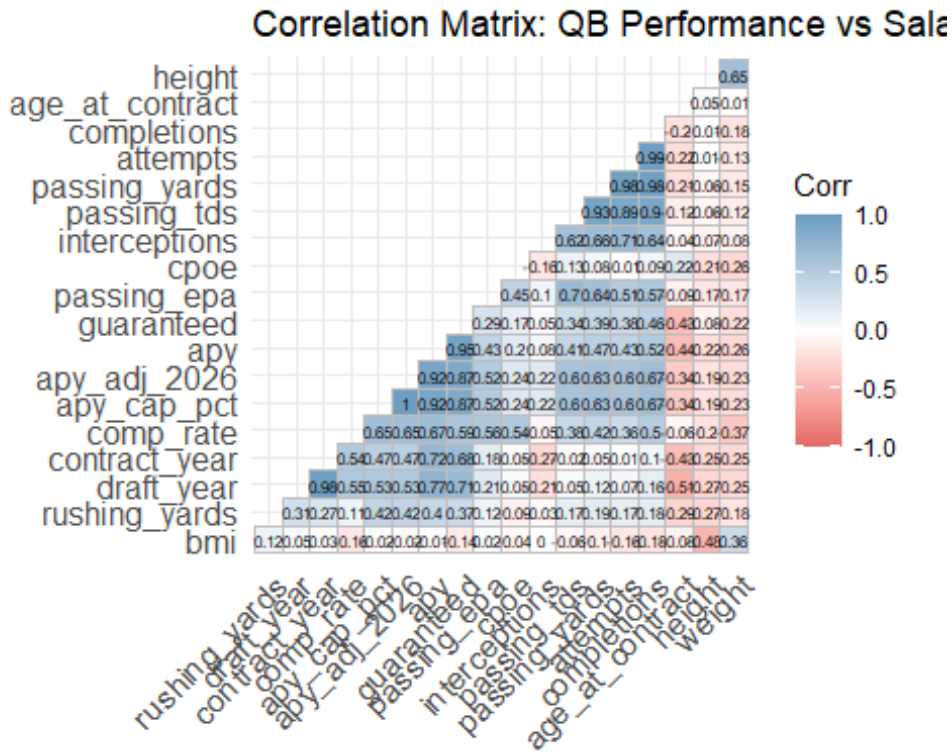
##
## Top Predictors for Adjusted APY (N = 39 ):
```

```r
print(sort(cor_matrix[,"apy_adj_2026"], decreasing = TRUE))
```

```
##     apy_cap_pct    apy_adj_2026             apy      guaranteed
completions
##      1.00000000      1.00000000      0.91979662      0.87027241
0.66544917
##       comp_rate   passing_yards         attempts     passing_tds
draft_year
##      0.64876171      0.62808145      0.59889139      0.59725669
0.52741990
##     passing_epa   contract_year   rushing_yards             cpoe
interceptions
##      0.51813260      0.46905084      0.41754508      0.23838103
0.22254536
##             bmi          height          weight age_at_contract
##     -0.02149371     -0.19075376     -0.22866855     -0.34372067
```

```r
ggcorrplot(cor_matrix, hc.order = TRUE, type = "lower", lab = TRUE, lab_size
= 2,
           method="square", colors = c("#E46764", "white", "#6D9EC1"),
           title="Correlation Matrix: QB Performance vs Salary",
ggtheme=theme_minimal())
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## ℹ Please use tidy evaluation idioms with `aes()`.
## ℹ See also `vignette("ggplot2-in-packages")` for more information.
## ℹ The deprecated feature was likely used in the ggcorrplot package.
##   Please report the issue at
<https://github.com/kassambara/ggcorrplot/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
#####################################################
# WIDE RECEIVCER EDA

wr_data <- read.csv("C:/R/STAT 7210/Project/wr_data_final.csv")

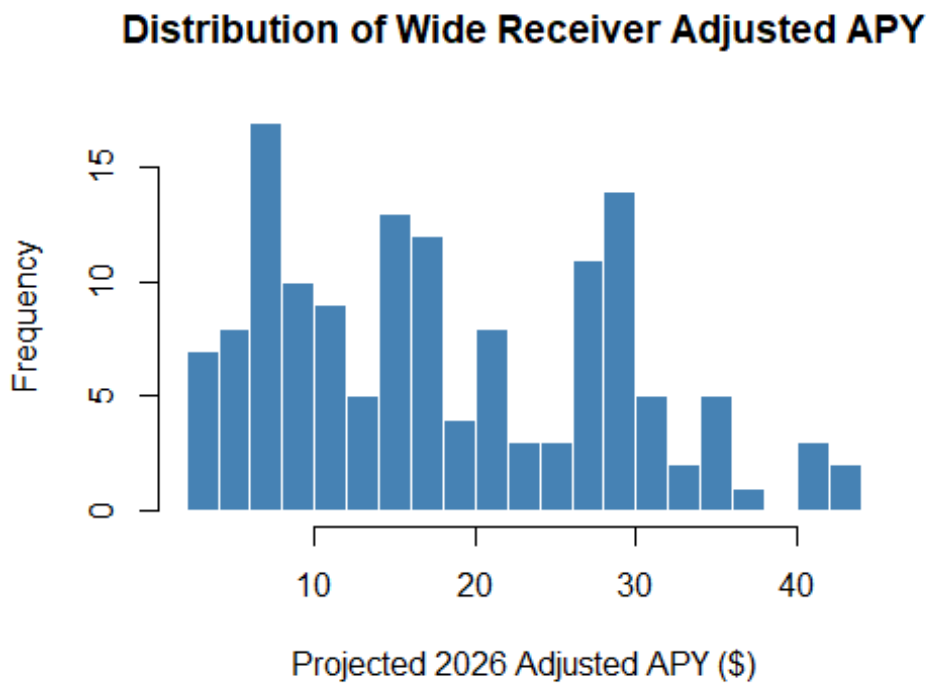
# Players below 3 million dollar salary are cutoff

wr_data <- wr_data |>
  filter(apy_adj_2026 >= 3.0) |>
  mutate(
    yards_per_target = receiving_yards / targets,
    adot = receiving_air_yards / targets,
    catch_rate = receptions / targets
  ) |>

  drop_na(targets, yards_per_target, adot, apy_adj_2026)

# TARGET VARIABLE HISTOGRAM

hist(wr_data$apy_adj_2026,
     main = "Distribution of Wide Receiver Adjusted APY",
     xlab = "Projected 2026 Adjusted APY ($)",
     col = "steelblue",      # Professional blue fill
     border = "white",       # Clean white borders between bars
     breaks = 15)            # Slightly more bins for the log-normal tail
```




```r
# 1x3 EDA boxplots

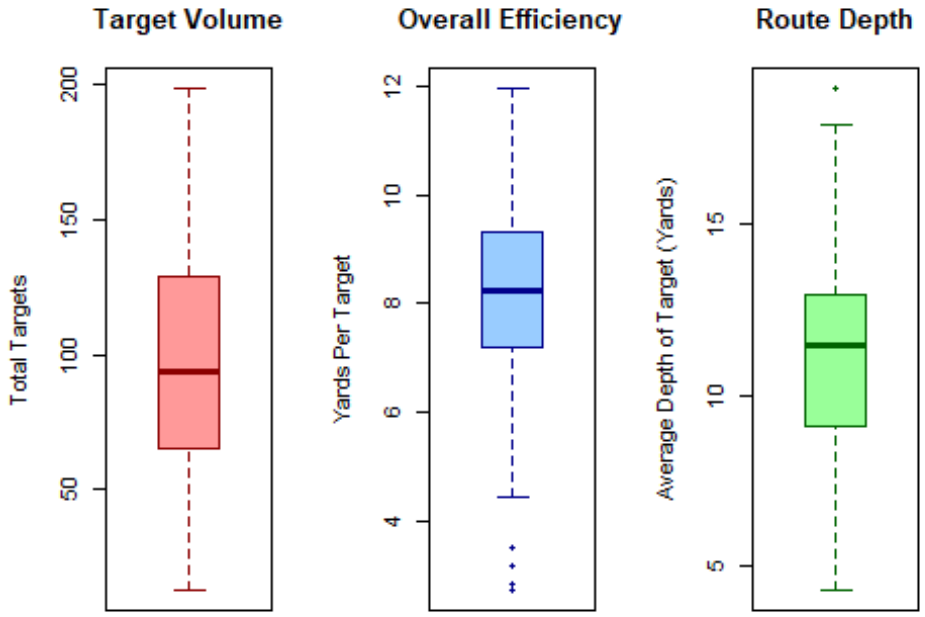
par(mfrow = c(1, 3))

#  Targets (Volume)
boxplot(wr_data$targets,
        main = "Target Volume",
        ylab = "Total Targets",
        col = "#ff9999",
        border = "darkred",
        pch = 16)

# Yards Per Target (Efficiency)
boxplot(wr_data$yards_per_target,
        main = "Overall Efficiency",
        ylab = "Yards Per Target",
        col = "#99ccff",
        border = "darkblue",
        pch = 16)

#  aDOT (Route Depth)
boxplot(wr_data$adot,
        main = "Route Depth",
        ylab = "Average Depth of Target (Yards)",
        col = "#99ff99",
```

```
        border = "darkgreen",
        pch = 16)
```




```
par(mfrow = c(1, 1))


# Correlatin Matrix

numeric_cols_wr <- wr_data |>
  select(where(is.numeric)) |>
  select(-any_of(c("player_id", "prior_year", "birth_year", "cap_value")))


numeric_cols_wr <- numeric_cols_wr[, colSums(is.na(numeric_cols_wr)) <
nrow(numeric_cols_wr)]
cor_matrix_wr <- cor(numeric_cols_wr, use = "pairwise.complete.obs")
cor_matrix_wr[is.na(cor_matrix_wr)] <- 0

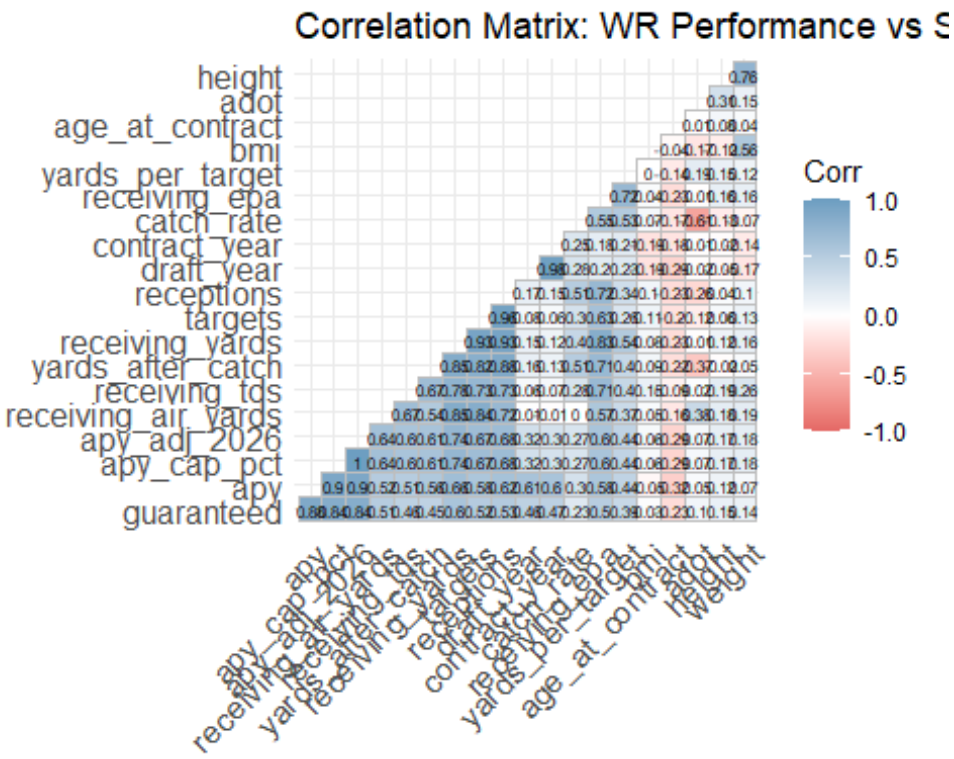

cat("\nTop Predictors for WR Adjusted APY (N =", nrow(wr_data), "):\n")

##
## Top Predictors for WR Adjusted APY (N = 142 ):

print(sort(cor_matrix_wr[,"apy_adj_2026"], decreasing = TRUE))
```

```
##         apy_cap_pct        apy_adj_2026                 apy
guaranteed
##          1.00000000          1.00000000          0.90403214
0.83548503
##     receiving_yards          receptions             targets
receiving_air_yards
##          0.73879376          0.67943561          0.67365224
0.64135457
##   yards_after_catch       receiving_epa       receiving_tds
yards_per_target
##          0.60622671          0.60443789          0.59940971
0.43606261
##          draft_year       contract_year          catch_rate
weight
##          0.31824187          0.30073972          0.26994402
0.17910919
##              height                adot                 bmi
age_at_contract
##          0.16537164          0.07052359          0.05704877         -
0.29481124
```

```
# Heatmap ggplot2
ggcorrplot(cor_matrix_wr, hc.order = TRUE, type = "lower", lab = TRUE,
lab_size = 2,
           method="square", colors = c("#E46764", "white", "#6D9EC1"),
           title="Correlation Matrix: WR Performance vs Salary",
ggtheme=theme_minimal())
```



```r
################################################################################
#####
# MODELING SECTION

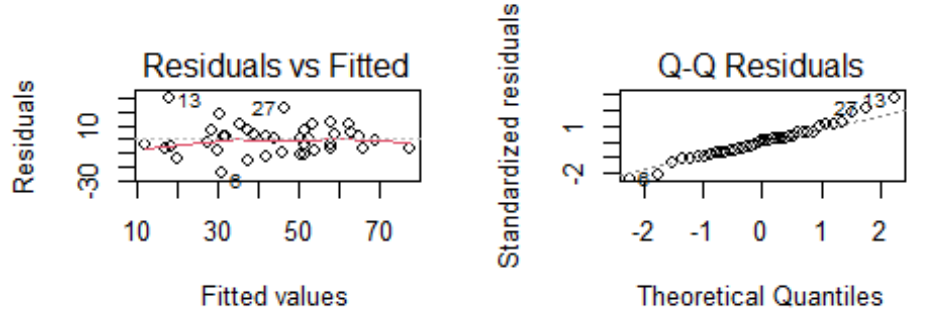


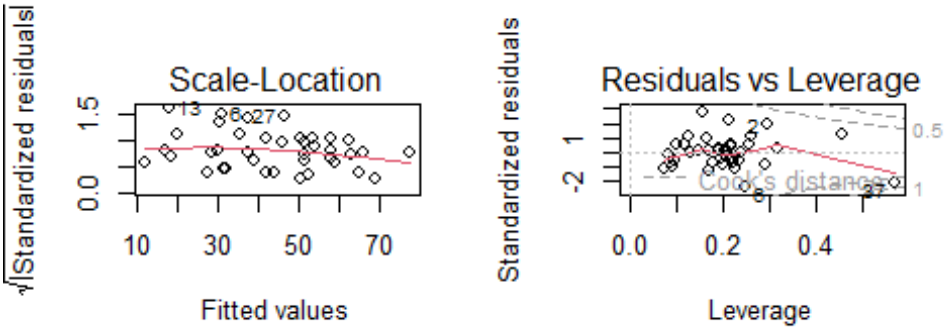
# QUARTERBACKS
# FULL UANDJUSTED MODEL
# DO NOT USE

model_full_qb <- lm(apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards +
                      interceptions + age_at_contract + cpoe + bmi,
                    data = qb_data)
summary(model_full_qb)

##
## Call:
## lm(formula = apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards +
##     interceptions + age_at_contract + cpoe + bmi, data = qb_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -23.849  -6.994   0.672   5.628  29.636
##
## Coefficients:
##                   Estimate Std. Error t value Pr(>|t|)
## (Intercept)     -41.823962  88.551423  -0.472  0.64001
## passing_tds       0.597966   0.300479   1.990  0.05547 .
## comp_rate         2.284759   0.616423   3.706  0.00082 ***
## rushing_yards     0.018712   0.008038   2.328  0.02662 *
## interceptions     0.236990   0.565669   0.419  0.67814
## age_at_contract  -3.640650   1.978488  -1.840  0.07534 .
## cpoe             -0.002714   0.634197  -0.004  0.99661
## bmi               0.560688   1.637836   0.342  0.73441
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.85 on 31 degrees of freedom
## Multiple R-squared:  0.695,  Adjusted R-squared:  0.6261
## F-statistic: 10.09 on 7 and 31 DF,  p-value: 1.623e-06

par(mfrow=c(2,2))
plot(model_full_qb)
```






```r
par(mfrow=c(1,1))


# FINAL MODEL

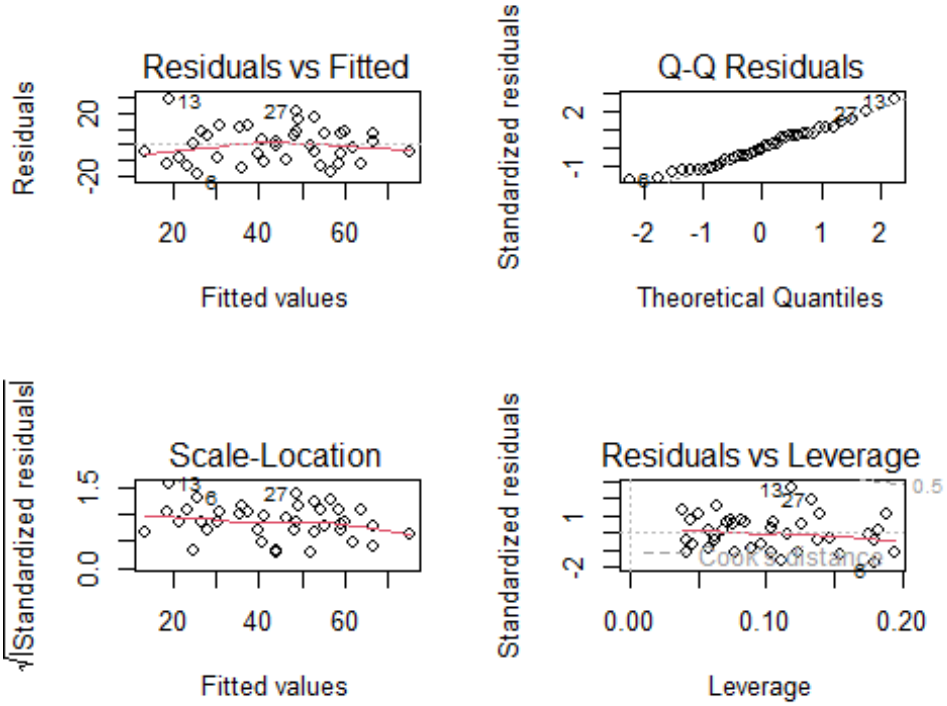
# inflation adjusted APY vs Passing Touchdowns + Completion Rate + Rushing
Yards + erors

model_final_qb <- lm(apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards,
data = qb_data)
summary(model_final_qb)

##
## Call:
## lm(formula = apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards,
##     data = qb_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -18.4375  -9.2565  -0.8214   7.9260  28.6189
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.160e+02  2.934e+01  -3.955 0.000356 ***
## passing_tds    7.131e-01  2.128e-01   3.351 0.001939 **
## comp_rate      2.170e+00  4.864e-01   4.461 8.07e-05 ***
```

```
## rushing_yards  2.266e-02  7.501e-03   3.021 0.004684 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.86 on 35 degrees of freedom
## Multiple R-squared:  0.655,  Adjusted R-squared:  0.6255
## F-statistic: 22.15 on 3 and 35 DF,  p-value: 3.225e-08
```

```
par(mfrow=c(2,2))
plot(model_final_qb)
```




```
par(mfrow=c(1,1))

# tests
anova(model_final_qb, model_full_qb)
```

```
## Analysis of Variance Table
##
## Model 1: apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards
## Model 2: apy_adj_2026 ~ passing_tds + comp_rate + rushing_yards +
interceptions +
##     age_at_contract + cpoe + bmi
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1     35 4926.7
## 2     31 4355.9  4    570.78 1.0155 0.4146
```

```r
# VIF Test
print(vif(model_final_qb))
```

```
##   passing_tds     comp_rate rushing_yards
##      1.192296      1.169979      1.032430
```

```r
# Shapiro Wilke Test for normality
print(shapiro.test(residuals(model_final_qb)))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(model_final_qb)
## W = 0.97125, p-value = 0.4099
```

```r
# Breusch Pagan Test for hetereoscedasticity
print(bptest(model_final_qb))
```

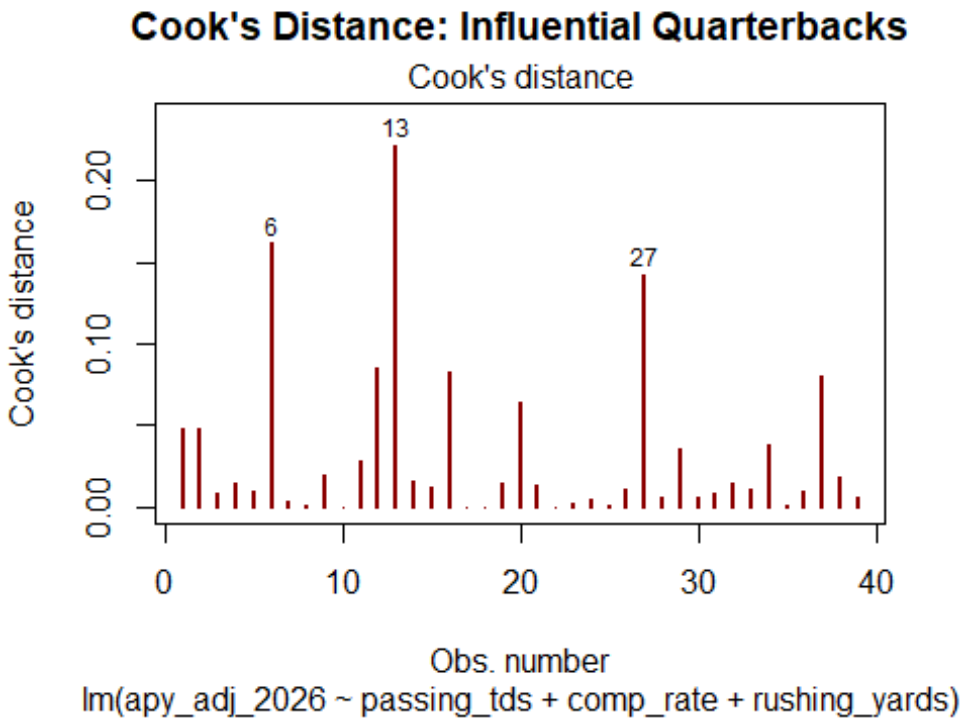
```
##
##  studentized Breusch-Pagan test
##
## data:  model_final_qb
## BP = 2.4259, df = 3, p-value = 0.4888
```

```r
#########################################################################

# Diagnostic plots 2x2
par(mfrow = c(2, 2))
plot(model_final_qb)
# Cook's D test
cooks_d <- cooks.distance(model_final_qb)
top_outliers <- sort(cooks_d, decreasing = TRUE)[1:3]
print(top_outliers)
```

```
##        13         6        27
## 0.2207501 0.1612151 0.1420102
```

```r
# Cook's D Plot
par(mfrow = c(1, 1))
plot(model_final_qb, which = 4,
     main = "Cook's Distance: Influential Quarterbacks",
     col = "darkred", lwd = 2)
```




```
###################################

# WIDE RECEIVER REGRESSION MODELING
###################################

wr_data <- read.csv("C:/R/STAT 7210/Project/wr_data_final.csv")

# 1. Apply Filters, Engineer Variables, and Drop NAs
wr_data <- wr_data |>
  filter(apy_adj_2026 >= 3.0) |>
  mutate(
    catch_rate = receptions / targets,
    adot = receiving_air_yards / targets,
    yac_per_reception = yards_after_catch / receptions,
    yards_per_target = receiving_yards / targets,
    td_per_target = receiving_tds / targets
  ) |>
  drop_na(targets, receptions, receiving_yards, receiving_tds,
          yards_after_catch, receiving_air_yards, receiving_epa,
          age_at_contract, bmi, catch_rate, adot, yac_per_reception,
          yards_per_target, td_per_target, apy_adj_2026)

# Full Model
model_full_log <- lm(log(apy_adj_2026) ~ targets + receptions +
receiving_yards +
                       receiving_tds + yards_after_catch +
```

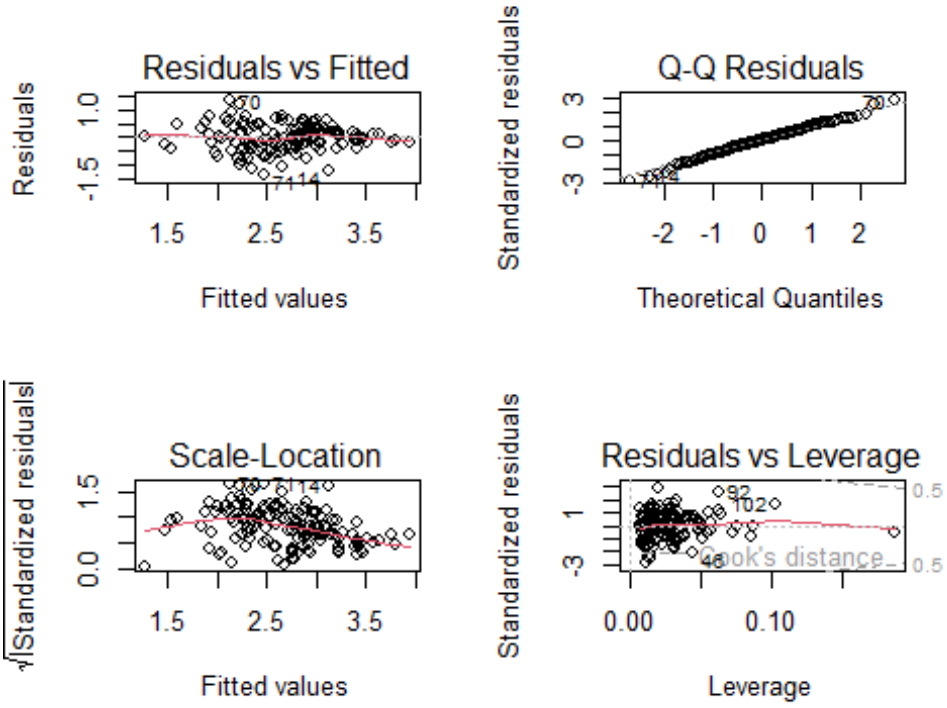
```
receiving_air_yards +
                        receiving_epa + age_at_contract + bmi + catch_rate +
adot +
                        yac_per_reception + yards_per_target + td_per_target,
                      data = wr_data)

# Final Reduced Model
model_reduced_log <- lm(log(apy_adj_2026) ~ targets + yards_per_target +
age_at_contract,
                         data = wr_data)

summary(model_reduced_log)

##
## Call:
## lm(formula = log(apy_adj_2026) ~ targets + yards_per_target +
##     age_at_contract, data = wr_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3038 -0.2735  0.0110  0.2956  1.2968
##
## Coefficients:
##                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.2918971  0.9583909   3.435 0.000783 ***
## targets           0.0085971  0.0009671   8.890 3.00e-15 ***
## yards_per_target  0.1140519  0.0239961   4.753 4.98e-06 ***
## age_at_contract  -0.0907856  0.0345620  -2.627 0.009595 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4691 on 138 degrees of freedom
## Multiple R-squared:  0.5308, Adjusted R-squared:  0.5206
## F-statistic: 52.03 on 3 and 138 DF,  p-value: < 2.2e-16

par(mfrow=c(2,2))
plot(model_reduced_log)
```




```r
par(mfrow=c(1,1))


# PArtial F-Test
anova(model_reduced_log, model_full_log)
```

```
## Analysis of Variance Table
##
## Model 1: log(apy_adj_2026) ~ targets + yards_per_target + age_at_contract
## Model 2: log(apy_adj_2026) ~ targets + receptions + receiving_yards +
##     receiving_tds + yards_after_catch + receiving_air_yards +
##     receiving_epa + age_at_contract + bmi + catch_rate + adot +
##     yac_per_reception + yards_per_target + td_per_target
##   Res.Df    RSS Df Sum of Sq      F Pr(>F)
## 1    138 30.363
## 2    127 29.058 11    1.3045 0.5183 0.8881
```

```r
##############################################################################333

# POST-HOC TESTS

# VIF Test
print(vif(model_reduced_log))
```

```
##          targets yards_per_target  age_at_contract
##         1.102204         1.080862         1.048837
```

```r
# Shapiro Wilke
print(shapiro.test(residuals(model_reduced_log)))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(model_reduced_log)
## W = 0.99393, p-value = 0.8141
```

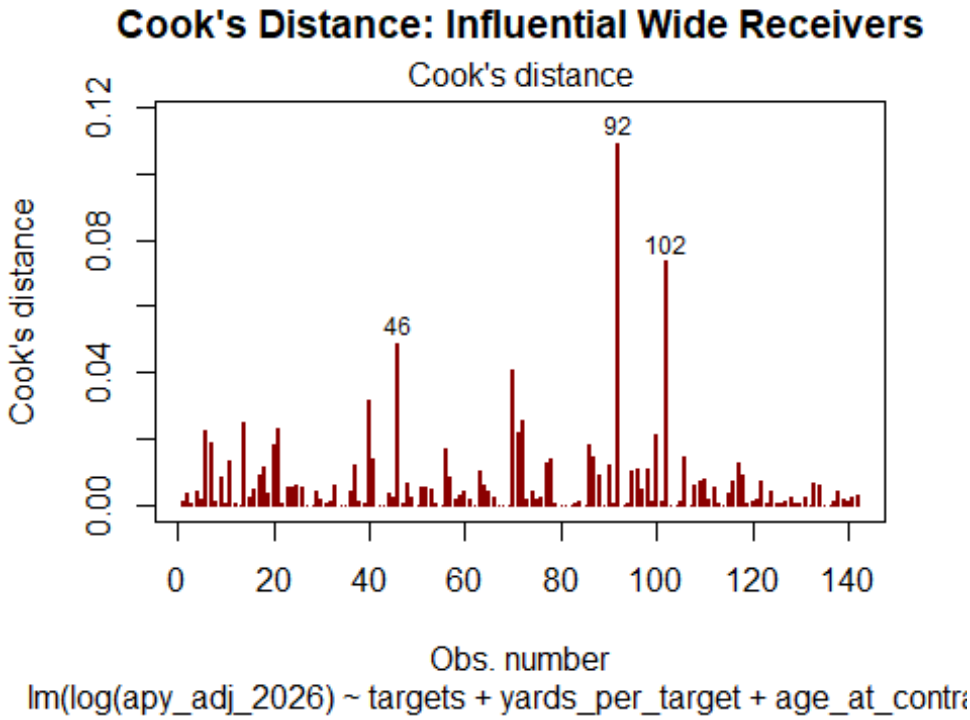
```r
# Breusch Pagan
print(bptest(model_reduced_log))
```

```
##
##  studentized Breusch-Pagan test
##
## data:  model_reduced_log
## BP = 11.868, df = 3, p-value = 0.007848
```

```r
# Diagnostic plots
par(mfrow = c(2, 2))
plot(model_reduced_log)
# Cook's D test
cooks_d_wr <- cooks.distance(model_reduced_log)
top_outliers_wr <- sort(cooks_d_wr, decreasing = TRUE)[1:3]
print(top_outliers_wr)
```

```
##         92        102         46
## 0.10880074 0.07344063 0.04866063
```

```r
# Cook's D Plot
par(mfrow = c(1, 1))
plot(model_reduced_log, which = 4,
     main = "Cook's Distance: Influential Wide Receivers",
     col = "darkred", lwd = 2)
```




```
# DURBIN WATSON TEST
durbinWatsonTest(model_reduced_log)
```

```
##  lag Autocorrelation D-W Statistic p-value
##    1      0.05279319      1.892657    0.52
##  Alternative hypothesis: rho != 0
```

```
#############################################################################
#################33
# POSTER VISUALISATIONS
#=====================================

# This whole section was made with Gemini AI as its just ggplot2
# Gemini is really good for making on the fly adjustments for your plots it
knows all the settings for
# geom line colours and hex codes. We don't need to do this manually

# Just visualisations, but feel
# free to check the code if you notice any mistake

# --- 1. RESIDUALS VS FITTED WITH SHADED ERROR REGIONS ---

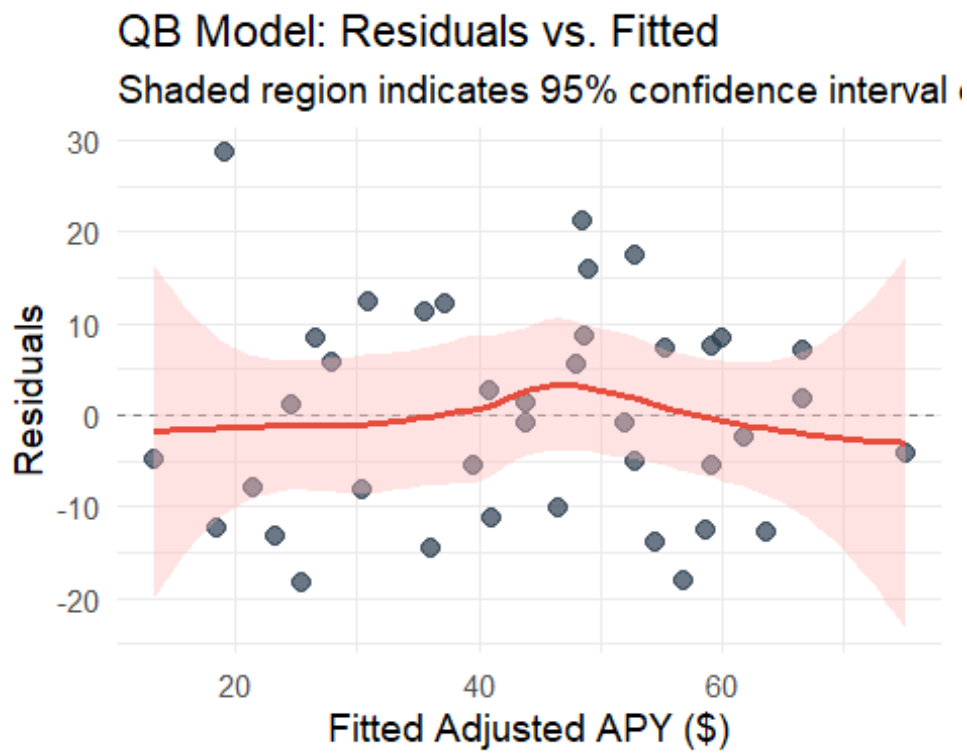
# QB Residuals Plot
qb_res_data <- data.frame(Fitted = fitted(model_final_qb), Residuals =
residuals(model_final_qb))
plot_qb_res <- ggplot(qb_res_data, aes(x = Fitted, y = Residuals)) +
```

```r
  geom_point(color = "#2c3e50", alpha = 0.7, size = 3) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_smooth(method = "loess", se = TRUE, color = "#e74c3c", fill =
"#ffbaba", alpha = 0.4) +
  labs(title = "QB Model: Residuals vs. Fitted",
       subtitle = "Shaded region indicates 95% confidence interval of the
variance",
       x = "Fitted Adjusted APY ($)", y = "Residuals") +
  theme_minimal(base_size = 14)

# WR Residuals Plot
wr_res_data <- data.frame(Fitted = fitted(model_reduced_log), Residuals =
residuals(model_reduced_log))
plot_wr_res <- ggplot(wr_res_data, aes(x = Fitted, y = Residuals)) +
  geom_point(color = "#2c3e50", alpha = 0.7, size = 3) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_smooth(method = "loess", se = TRUE, color = "#e74c3c", fill =
"#ffbaba", alpha = 0.4) +
  labs(title = "WR Model: Residuals vs. Fitted",
       subtitle = "Shaded region indicates 95% confidence interval of the
variance",
       x = "Fitted Log(Adjusted APY)", y = "Residuals") +
  theme_minimal(base_size = 14)

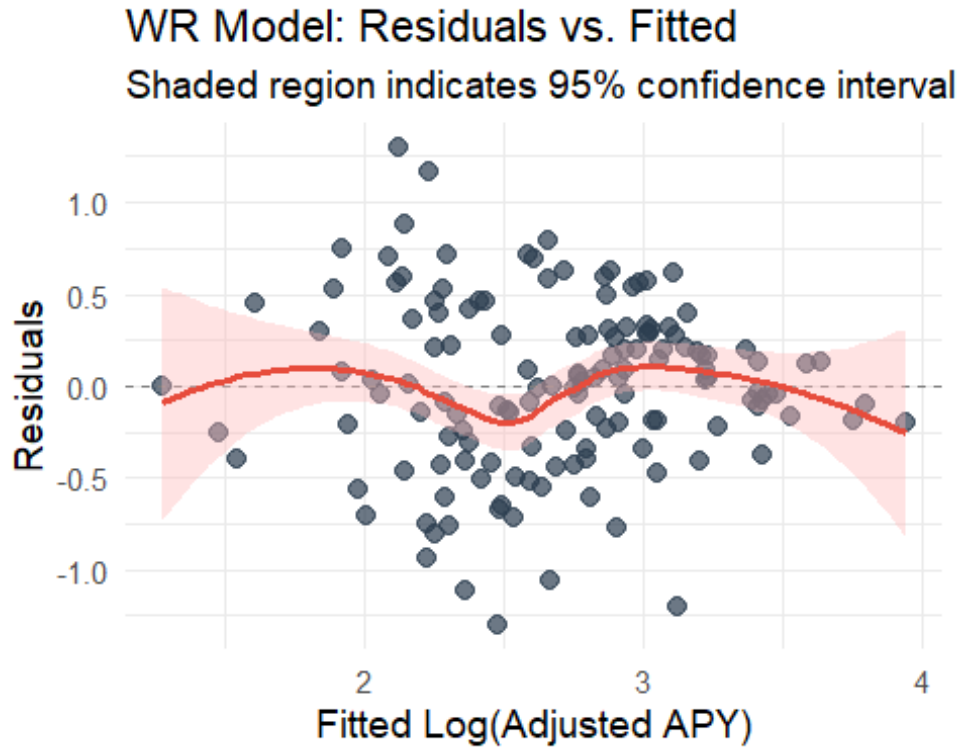
print(plot_qb_res)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
print(plot_wr_res)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

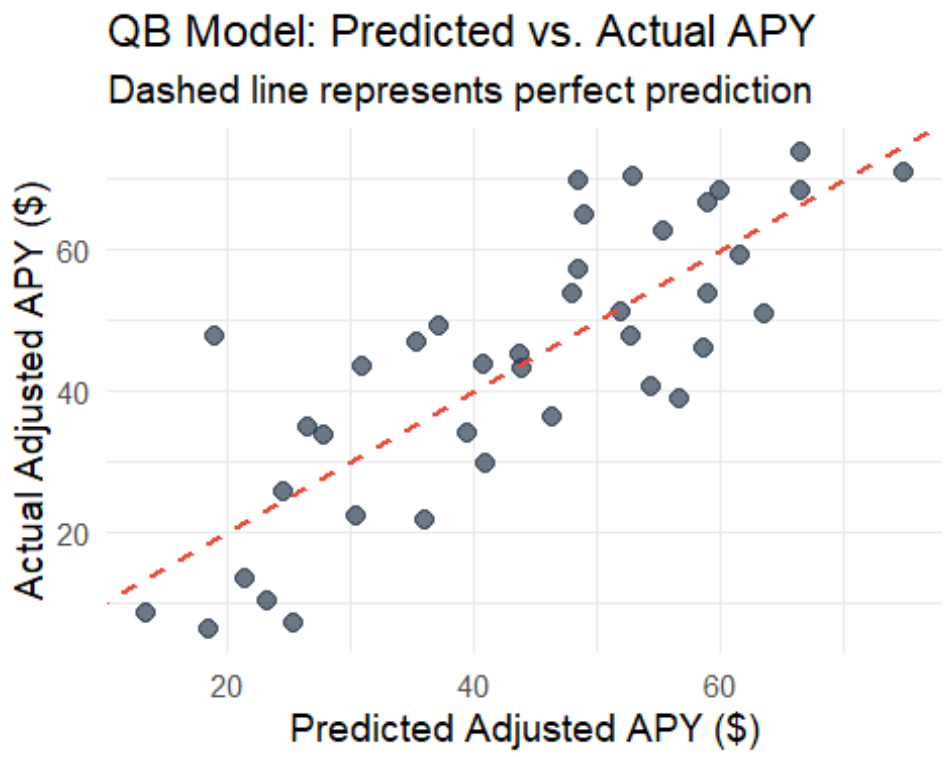


```r
# PREDICTED VS ACTUAL (accuracy plot)

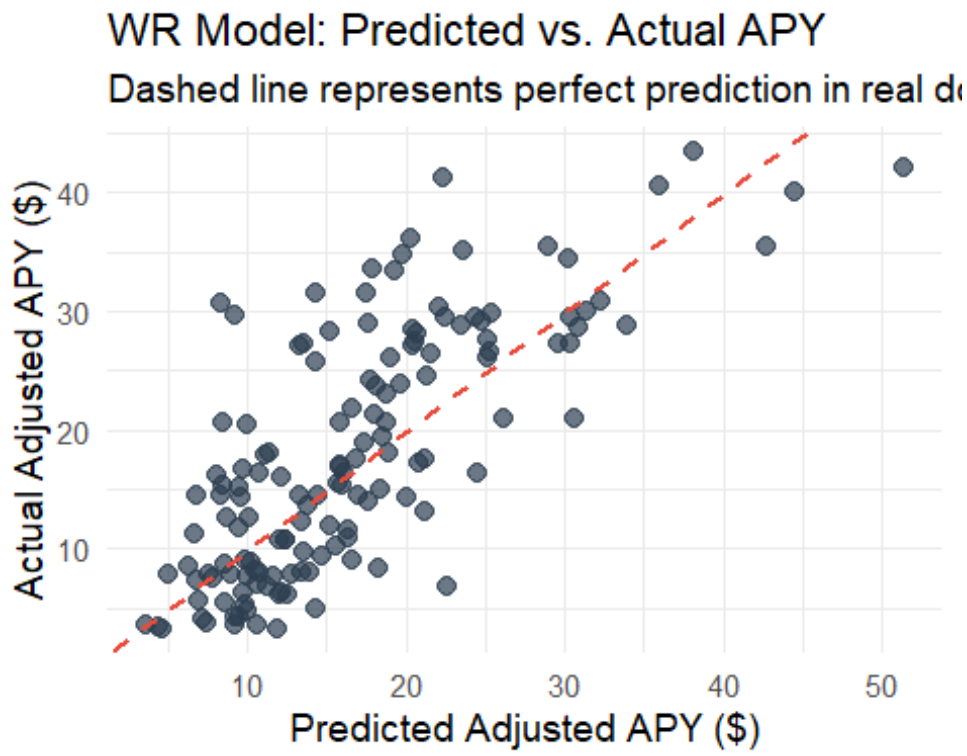
# QB Pred vs Actual
pred_data_qb <- data.frame(Actual = qb_data$apy_adj_2026, Predicted =
fitted(model_final_qb))
plot_qb_pred <- ggplot(pred_data_qb, aes(x = Predicted, y = Actual)) +
  geom_point(color = "#2c3e50", alpha = 0.7, size = 3) +
  geom_abline(intercept = 0, slope = 1, color = "#e74c3c", linetype =
"dashed", linewidth = 1) +
  labs(title = "QB Model: Predicted vs. Actual APY",
       subtitle = "Dashed line represents perfect prediction",
       x = "Predicted Adjusted APY ($)", y = "Actual Adjusted APY ($)") +
  theme_minimal(base_size = 14)

# WR Pred vs Actual (transofmred back to real dollars too)
pred_data_wr <- data.frame(Actual = wr_data$apy_adj_2026, Predicted =
exp(fitted(model_reduced_log)))
plot_wr_pred <- ggplot(pred_data_wr, aes(x = Predicted, y = Actual)) +
  geom_point(color = "#2c3e50", alpha = 0.7, size = 3) +
  geom_abline(intercept = 0, slope = 1, color = "#e74c3c", linetype =
"dashed", linewidth = 1) +
  labs(title = "WR Model: Predicted vs. Actual APY",
       subtitle = "Dashed line represents perfect prediction in real
dollars",
       x = "Predicted Adjusted APY ($)", y = "Actual Adjusted APY ($)") +
  theme_minimal(base_size = 14)

print(plot_qb_pred)
```



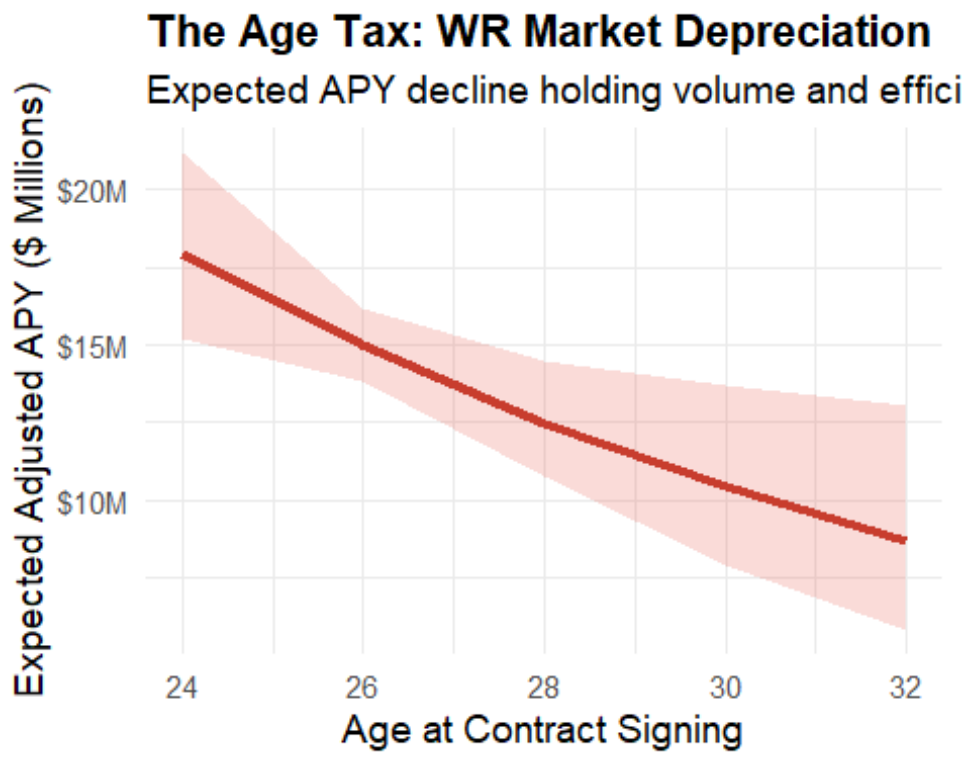

```
print(plot_wr_pred)
```



```r
# Age Tax negative line graph for Wide Receivers

age_effect <- effect("age_at_contract", model_reduced_log)
age_effect_df <- as.data.frame(age_effect)
# back transform  to get it in USD again
age_effect_df$fit_dollars <- exp(age_effect_df$fit)
age_effect_df$lower_dollars <- exp(age_effect_df$lower)
age_effect_df$upper_dollars <- exp(age_effect_df$upper)

# Visualisation
ggplot(age_effect_df, aes(x = age_at_contract, y = fit_dollars)) +
  geom_line(color = "#c0392b", linewidth = 1.5) +
  geom_ribbon(aes(ymin = lower_dollars, ymax = upper_dollars), alpha = 0.2,
fill = "#e74c3c") +
  labs(title = "The Age Tax: WR Market Depreciation",
       subtitle = "Expected APY decline holding volume and efficiency
constant",
       x = "Age at Contract Signing",
       y = "Expected Adjusted APY ($ Millions)") +
  scale_y_continuous(labels = scales::dollar_format(scale = 1, suffix = "M"))
+
  theme_minimal(base_size = 14) +
  theme(plot.title = element_text(face = "bold"))
```




```r
##############################################################################
#################################3
```

```
# SANDBOX (Discarded Code)


# # Coefficient Forest plots (unused)
#
# # QB Forest Plot
# coefs_qb <- as.data.frame(summary(model_final_qb)$coefficients)
# ci_qb <- as.data.frame(confint(model_final_qb))
# forest_data_qb <- cbind(coefs_qb, ci_qb)[-1, ]
# forest_data_qb$Term <- c("Passing TDs", "Completion Rate", "Rushing Yards")
#
# plot_qb_forest <- ggplot(forest_data_qb, aes(x = Estimate, y =
reorder(Term, Estimate))) +
#   geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
#   geom_errorbarh(aes(xmin = `2.5 %`, xmax = `97.5 %`), height = 0.2, color
= "#2980b9", linewidth = 1) +
#   geom_point(size = 4, color = "#c0392b") +
#   labs(title = "QB Effect Sizes (Beta Coefficients)",
#        subtitle = "Error bars represent 95% Confidence Intervals",
#        x = "Estimated Impact on Adjusted APY ($)", y = "") +
#   theme_minimal(base_size = 14)
#
# # WR Forest Plot

# coefs_wr <- as.data.frame(summary(model_reduced_log)$coefficients)
# ci_wr <- as.data.frame(confint(model_reduced_log))
# forest_data_wr <- cbind(coefs_wr, ci_wr)[-1, ]
# forest_data_wr$Term <- c("Targets", "Targets Squared", "Yards Per Target")
#
# plot_wr_forest <- ggplot(forest_data_wr, aes(x = Estimate, y =
reorder(Term, Estimate))) +
#   geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
#   geom_errorbarh(aes(xmin = `2.5 %`, xmax = `97.5 %`), height = 0.1, color
= "#2980b9", linewidth = 1) +
#   geom_point(size = 4, color = "#c0392b") +
#   labs(title = "WR Effect Sizes (Log-Linear Coefficients)",
#        subtitle = "Impact on the Natural Log of Adjusted APY",
#        x = "Estimated Beta Coefficient", y = "") +
#   theme_minimal(base_size = 14)
#
# print(plot_qb_forest)
# print(plot_wr_forest)
#
#

# model_test_1 <- lm(log(apy_adj_2026) ~ log(targets) + yards_per_target,
data = wr_data)
# autoplot(model_test_1)
```

```
# model_test_2 <- lm(log(apy_adj_2026) ~ targets + I(targets^2) +
yards_per_target, data = wr_data)
# autoplot(model_test_2)
# model_test_3 <- lm(log(apy_adj_2026) ~ targets * yards_per_target, data =
wr_data)
# autoplot(model_test_3)

#
# model_log_log <- lm(log(apy_adj_2026) ~ log(targets) + yards_per_target,
data = wr_data)
# summary(model_log_log)
# autoplot(model_log_log)
#
#
# model_interaction <- lm(log(apy_adj_2026) ~ targets * yards_per_target,
data = wr_data)
# summary(model_interaction)
# autoplot(model_interaction)
#
#
# model_test_age <- lm(log(apy_adj_2026) ~ targets + yards_per_target +
age_at_contract, data = wr_data)
# summary(model_test_age)
# autoplot(model_test_age, which = 1)
#
#




# durbinWatsonTest(model_final_qb)
#
# # Pull the residual for Observation 27 (Lamar Jackson)
# qb_residuals <- residuals(model_final_qb)
# print(qb_residuals[27])
#
#
#
#
#
# # WR sqr rt model
# model_balance <- lm(log(apy_adj_2026) ~ sqrt(targets) + yards_per_target +
age_at_contract,
#                         data = wr_data)
#

# summary(model_balance)

# gold_res_data <- data.frame(
#   Fitted = fitted(model_balance),
#   Residuals = residuals(model_balance)
```

```
# )
#

# library(ggplot2)
# ggplot(gold_res_data, aes(x = Fitted, y = Residuals)) +
#   geom_point(color = "#2c3e50", alpha = 0.7, size = 3) +
#   geom_hline(yintercept = 0, linetype = "dashed", color = "black",
linewidth = 1) +
#   geom_smooth(method = "loess", se = TRUE, color = "#e74c3c", fill =
"#ffbaba", alpha = 0.4) +
#   labs(title = "WR 'Goldilocks' Model: Residuals vs. Fitted",
#        subtitle = "Testing Sqrt(Targets) for optimal parsimony",
#        x = "Fitted Log(Adjusted APY)",
#        y = "Residuals") +
#   theme_minimal(base_size = 14)
#

# plot(model_goldilocks, which = 1,2)
#
#
#
#
#
# # Spline attempts
# library(splines)
# library(car)
#
# # Spline Model Test (failed)
# # ================================================================
# cat("\n--- VARIATION 1: Target Spline (df = 3) ---\n")
# model_spline_1 <- lm(log(apy_adj_2026) ~ ns(targets, df = 3) +
yards_per_target + age_at_contract,
#                      data = wr_data)
#
# print(summary(model_spline_1))
# print(vif(model_spline_1))
#
# par(mfrow = c(2, 2))
# plot(model_spline_1, main = "Var 1: Targets df=3")
# par(mfrow = c(1, 1))
#
```